* Revision:

* feature Engineering:

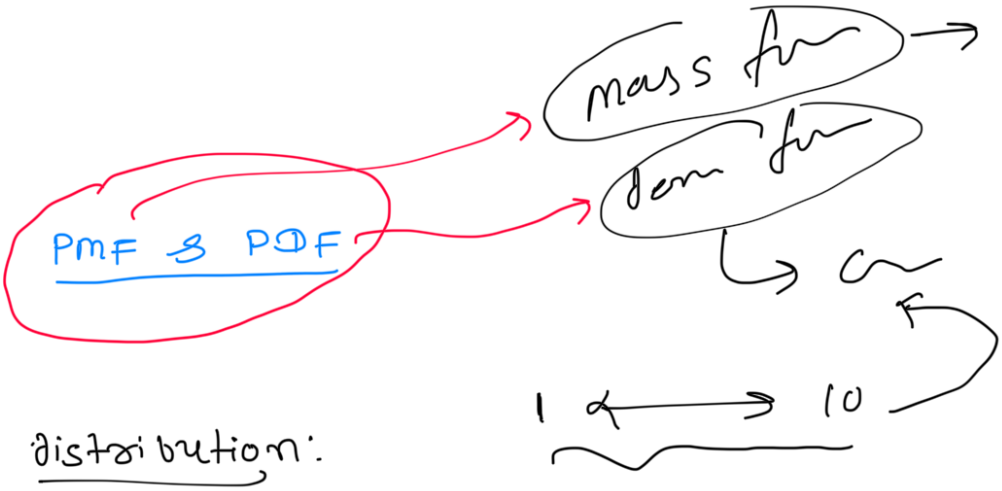
→ Category due

→ Pending: Transformers & feature Scaling

& z-score

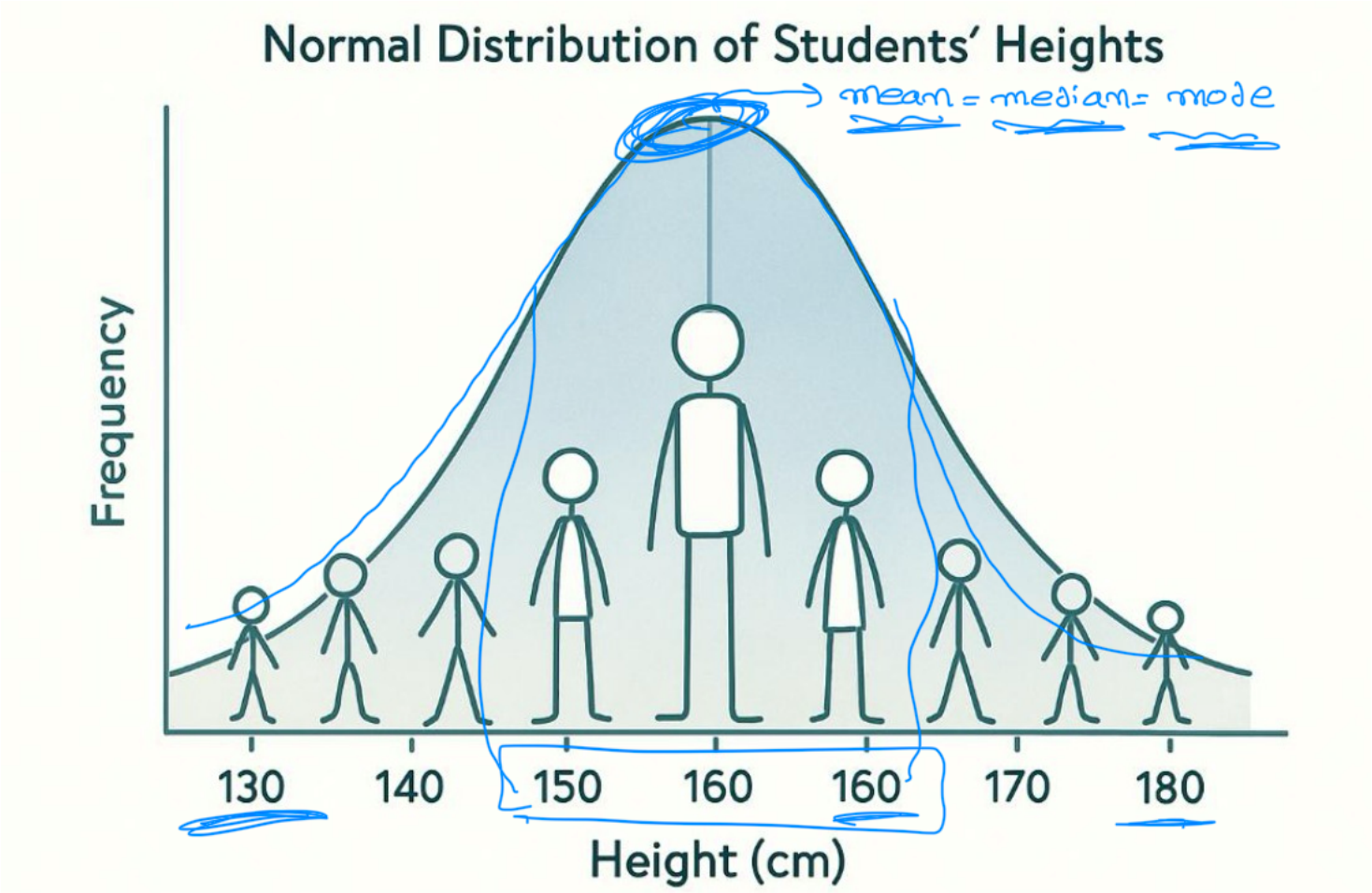
* Stistics:

- L12/L13 : PMF & PDF



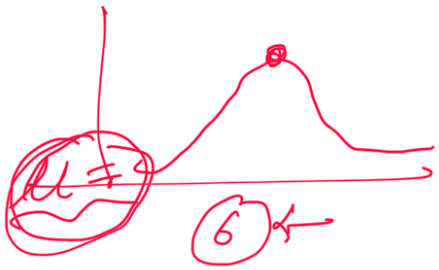
① Normal distribution:

- Bell shaped curve & symmetric around mean.


Normal Distribution of Students' Heights
mean = median = mode
Frequency
130
140
150
160
160
170
180
Height (cm)

desmos.com





$$\mu = \frac{0+5+10+15+20+25+30}{7} = 15$$

$\sigma = 10$

$25 - 15 = 10$

1 std for

$30 - 15 = 15$

$= 15$

σ: spread 0 5 10 15 20 25 30

$$\sigma^2 = \frac{\Sigma (x-\mu)^2}{n} = \frac{(0-15)^2 + (5-15)^2 + (10-15)^2 + (15-15)^2 + (20-15)^2 + (25-15)^2 + (30-15)^2}{7}$$

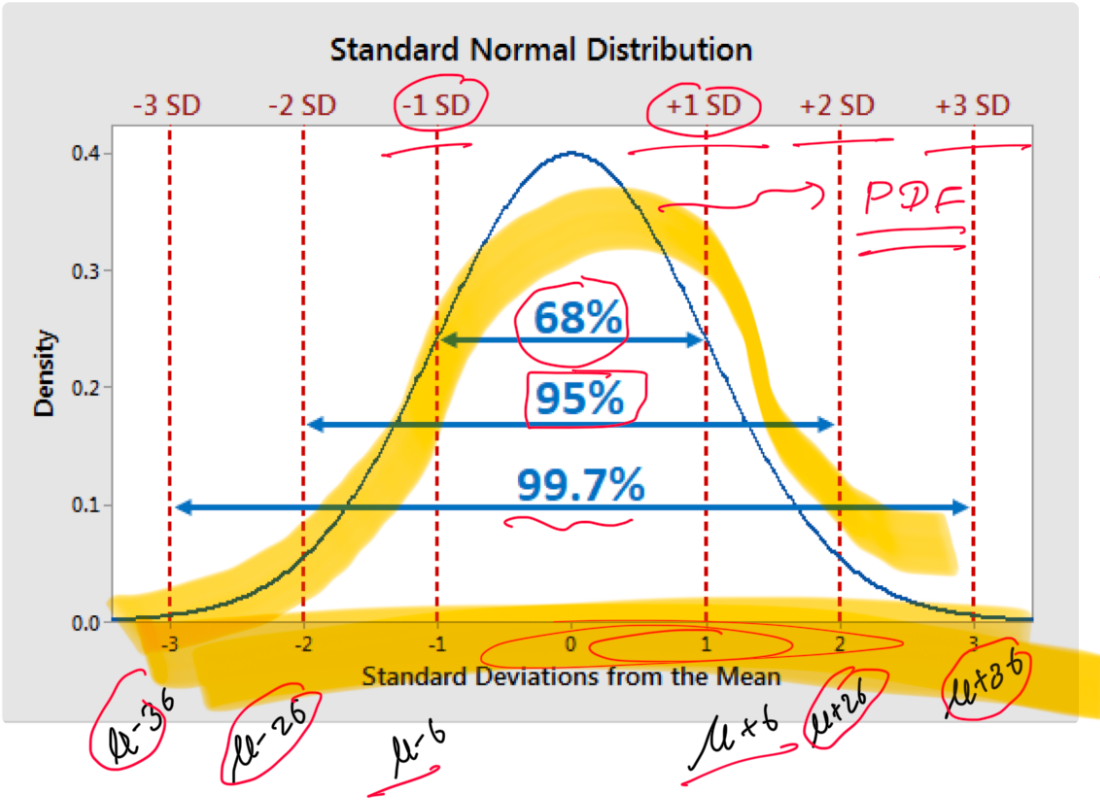
$$= \frac{(-15)^2 + (-10)^2 + (-5)^2 + 0 + (5)^2 + (10)^2 + (15)^2}{7}$$

$$= \frac{225+100+25+25+100+225}{7}$$

$$\sigma^2 = \frac{700}{7} = 100$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{100} = 10$$

* Empirical Rule:



PDF

Dm

→ If we know μ & σ of any dataset & we also know it is normally distributed then we can plot the distribution.

→

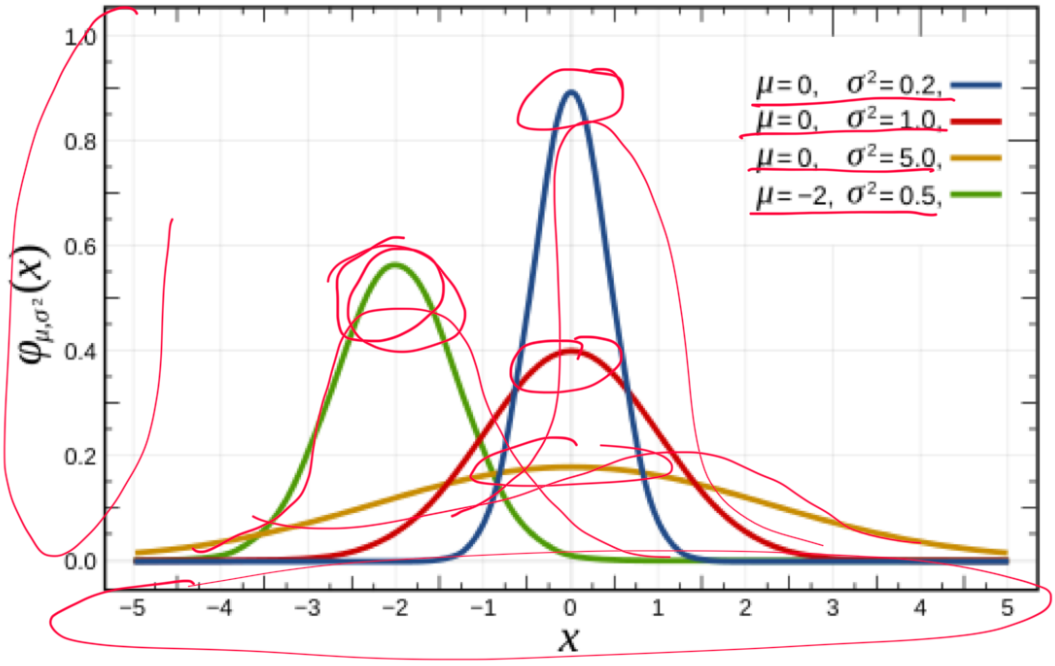
$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2}$$

⇒ Properties of Normal distribution:

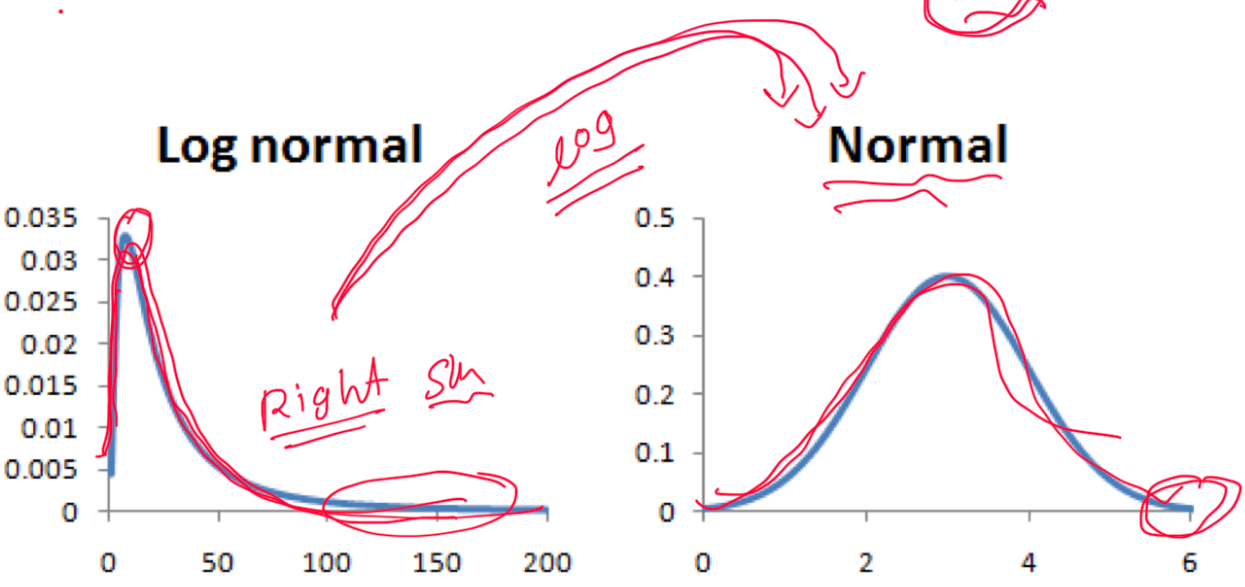
- Symmetric around it's mean.
- mean = median = mode
- Empirical Rule
- Area under the curve = 1 (?)
- parameters: (μ, σ)

$X \sim N(5, 10)$

$X \sim N(\mu, \sigma)$



* Log Normal distribution:



* Prob. dis. whose logarithm is normally distributed.

$X \sim \log\text{normal}(\mu, \sigma)$

$\log(X) \sim N(\mu', \sigma')$

Eg: Income of the people.

Usecase: Convert Right skewed data → Normal dis. & outlier

* Z score:

- How many std a data point lies away from the mean.

Eg: 0 5 10 15 20 25 30 → 15

1 +0.5

→ 6=10 → 1.5 → -5

x=35

μ=15
6=10
f=30 } 1.5 std

40

40 - 15 = 25 Unit

6=10 → 25/10 = 2.5 std

x=35 ? → 2 std

35 - 15 = 20

(20)/6 => 20/10 = 2 std

x = 40 : x - μ => 40 - 15 = 25/6 = 25/10 = 2.5 std

→ x = -5 → (-5 - 15)/10 = -20/10 = -2 std

5 & 6

3.53

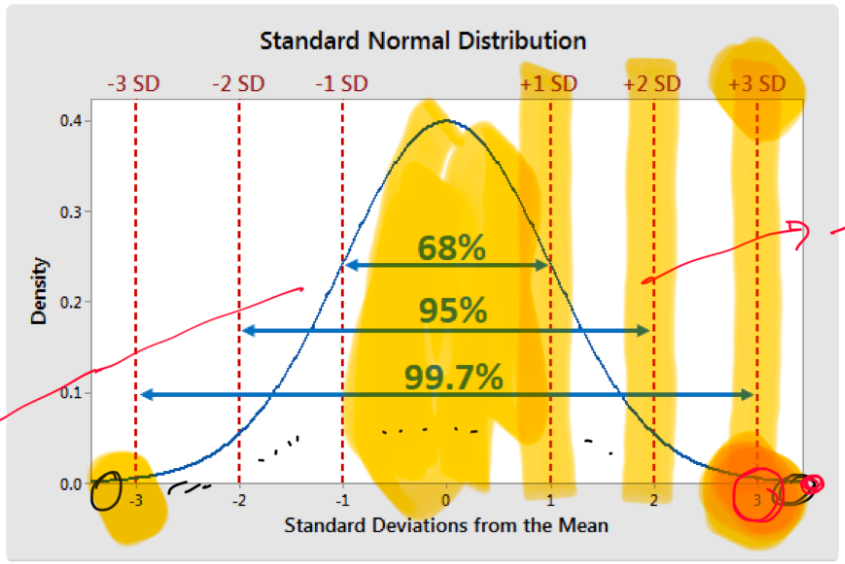
Schm Tm

→ 2 sam

→ z score = $\frac{x - \mu}{\sigma}$ z score of μ = 0

* what can we know from z score ?

-



→ If |z| > 3 → very less proba

z = 0.5 → high proba

→ we can compare diff. data points.

* z-score method to detect outliers :

if |z| > 3 → outliers

UPA } z score

* Standard Normal distribution:

- Normal dist. with $\mu = 0$ & $\sigma = 1$.

Unit z-score

Suppose two colleges conduct different exams. We want to compare the performance of two students.

Details	College A	College B
Class average (mean)	70	50
Standard deviation	10	5
Student's marks	85	60

Question: Which student performed better compared with their own classmates?

1.5 ← 2

→ A:

$$z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$$

B:

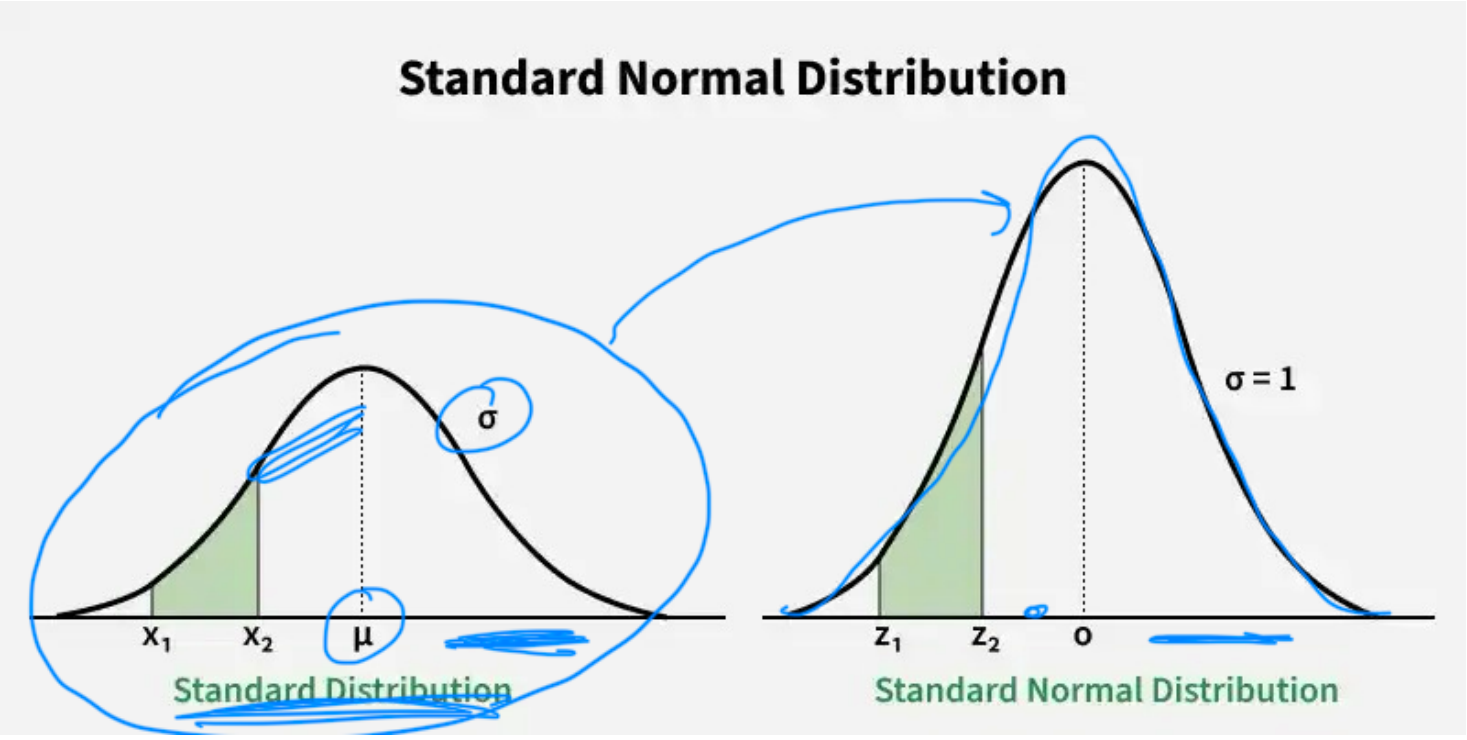
$$z = \frac{x - \mu}{\sigma} = \frac{60 - 50}{5} = \frac{10}{5} = 2$$

"Raw marks alone may not tell us how well a student performed relative to their classmates. By converting marks into z-scores, we measure how far each student is from their own class average in standard deviation units."

→ Convert og value → Z-score } For Normally distributed data

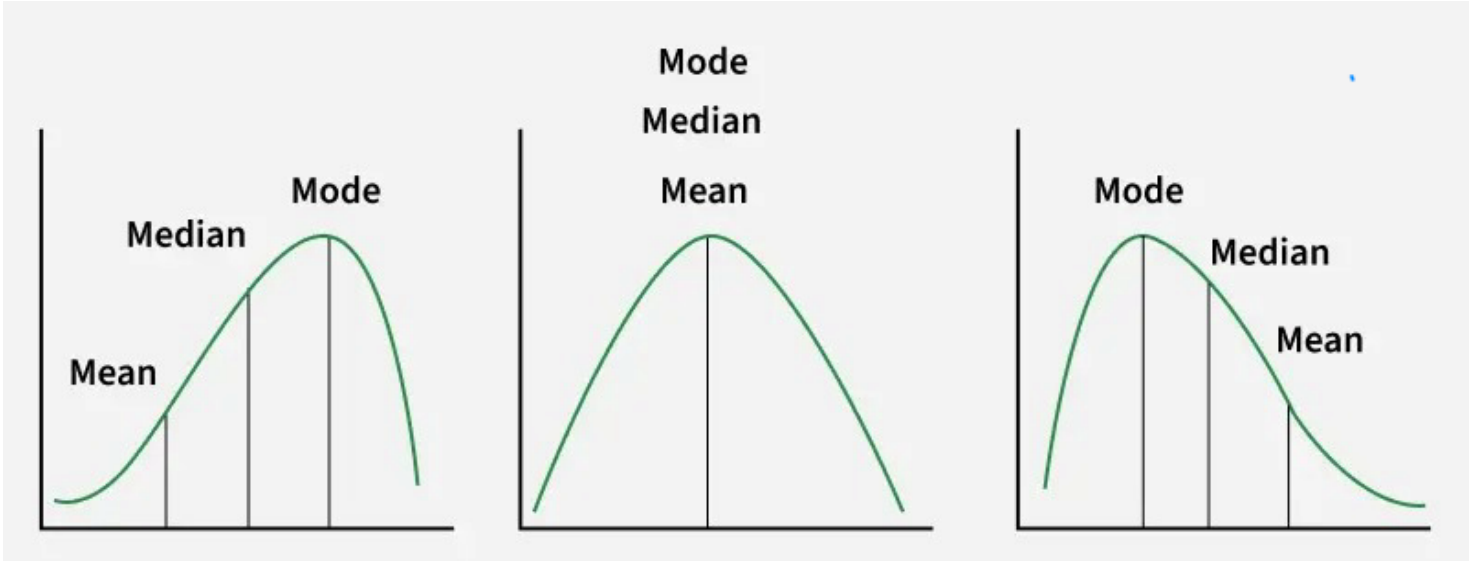
it gives $\mu = 0, \sigma = 1$ } std Normn dis.

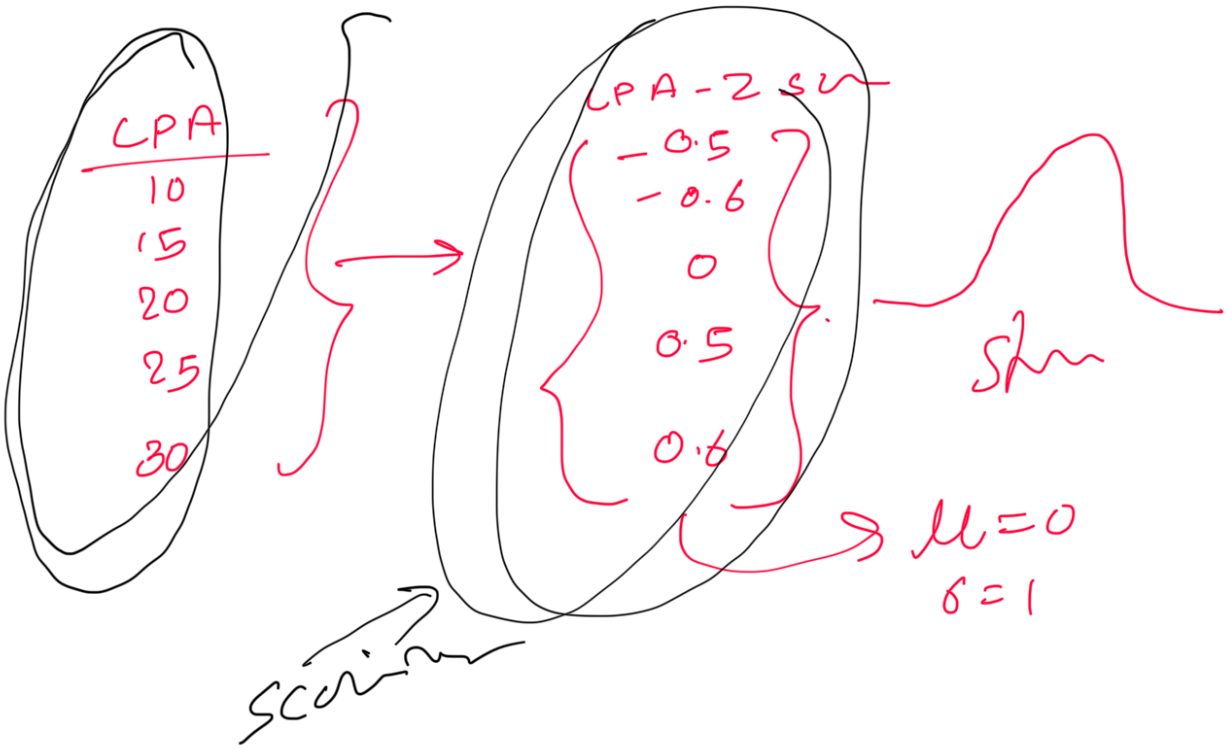
$X \sim N(0, 1)$



→ we can convert diff. dist. to the same standard scale for comparision.

* Skewness:


Mean
Median
Mode
Mode
Median
Mean
Mode
Median
Mean



$$\frac{x-\mu}{\sigma} \rightarrow \mu = 0,\ \sigma = 1$$

$$y = m_1 x_1 + m_2 x_2 + c$$

x_1 (Income) x_2 (Height Feet)

10,000 0	5.5
12,00	56

Unit = same

Z-score



mode

000